

NETWORK SECURITY

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Table of Content

- 1. Objective**
- 2. Network Architecture**
- 3. IP Addressing & Segmentation Strategy**
- 4. Technical Implementation & Security Configurations**
- 5. Layer 2 Threat Analysis & Mitigation**
- 6. Testing, Verification & Live Demo**
- 7. Project Conclusion & Q&A**

1. Objective

- Design a secure enterprise network for ABC Tech Solutions
- Ensure segmentation using VLANs (10, 20, 30)
- Implement controlled communication between departments
- Provide secure internet access using ASA Firewall (PAT)
- Enforce Guest network isolation
- Enable secure remote access using Site-to-Site VPN

2. Network architecture

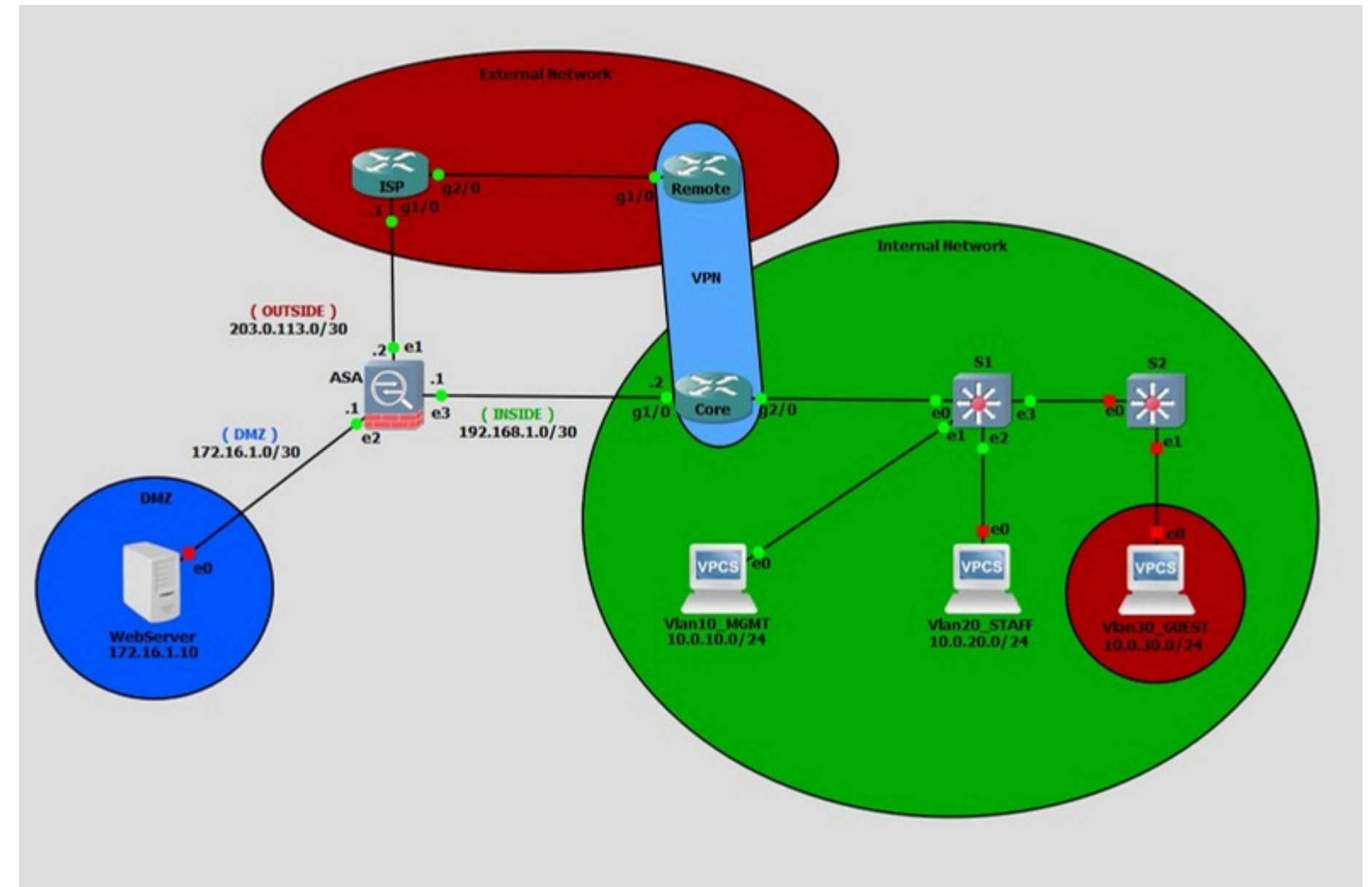
Core Router connects Switches (S1, S2) and ASA Firewall

ASA separates:

Inside (LAN)
DMZ (Public Server)
Outside (ISP)

Traffic Flow:

- VLAN → Core Router → ASA → Internet
- Remote Site → VPN → Core → Internal VLANs
- Router-on-a-Stick handles inter-VLAN routing



3. IP Addressing & Segmentation Strategy

The network uses Router-on-a-Stick configuration on the Core Router's g2/0 interface:

- **VLAN 10 (MGMT)**: Administrative control
Can SSH to ASA
- **VLAN 20 (STAFF)**: Employee operations.
- **VLAN 30 (GUEST)**: Restricted to other VLAN but can access outside .(DHCP)

Device	Interface	IP	Subnet mask
ASA	e1/0	203.0.113.2	255.255.255.252
ASA	e2/o	172.16.1.1	255.255.255.252
ASA	e3/0	192.168.1.1	255.255.255.252
Core	g1/0	192.168.1.2	255.255.255.252
Core	g2/0	N/A	N/A
Core	g2/0.10	10.0.10.1	255.255.255.0
Core	g2/0.20	10.0.10.2	255.255.255.0
Core	g2/0.30	10.0.10.3	255.255.255.0
DMZ	e0	172.16.0.10	255.255.255.252
ISP	g1/0	203.0.113.1	255.255.255.252
ISP	g2/0	198.51.100.1	255.255.255.252
Remote(R)	g1/0	198.51.100.2	255.255.255.252

4. Technical Implementation & Security Configurations

Device Hardening:

- Disable unused services: no http, no dns lookup
- Enforce strong authentication: encrypted passwords, local users
- Secure remote access: SSH v2 only, no Telnet
- Shutdown unused ports: prevent unauthorized access
- Apply firewall protections (ASA): (security levels, threat detection)

Layer 2 (Switching)

- VLANs created to segment network:
VLAN 10 → Management
VLAN 20 → Staff
VLAN 30 → Guest
- 802.1Q Trunking
Enables multiple VLANs between:
Switch ↔ Switch
Switch ↔ Core Router
- Access Ports
Assigned to specific VLANs
spanning-tree portfast for fast device connection

4. Technical Implementation & Security Configurations (cont.)

Routing:

- Router-on-a-Stick Subinterfaces on Core Router:

G2/0.10 → 10.0.10.1

G2/0.20 → 10.0.20.1

G2/0.30 → 10.0.30.1

- Enables inter-VLAN communication
- Static Routing Core → ASA (default route)
- ASA → ISP (internet access)
- ISP routes back to internal network

Access Control Lists (ACLs):

- Control traffic between VLANs
- Enforce least privilege principle
- Protect sensitive resources

VLAN	Rule
VLAN 10	Permit 10.0.10.0 → 10.0.20.0, Deny → VLAN 30
VLAN 20	Permit 10.0.20.0 → 10.0.10.0, Deny → VLAN 30
VLAN 30	Deny → VLAN 10 & 20, Permit → Internet

4. Technical Implementation & Security Configurations (cont.)

Firewall & NAT:

- Control and filter traffic between trusted and untrusted networks
- Protect internal resources from external threats
- Enable internet access for internal users

Secure Remote Access:

- Protect administrative access to network devices
- Ensure only authorized personnel can manage routers, switches, and ASA
- Encrypt all remote management sessions

VPN (Site-to-Site IPsec)

- Secure tunnel between:
Core Router ↔ Remote Router
- Security Features
Encryption: **AES**
Authentication: **Pre-shared key**
Integrity: **SHA**
- Traffic Encryption
Only defined traffic is protected:
10.0.0.0 ↔ 192.168.100.0

5. Layer 2 Threat Analysis & Mitigation

Common Layer 2 Threats

ARP Spoofing (MITM Attack)

- Attacker sends fake ARP replies
- Redirects traffic through attacker

DHCP Spoofing

- Rogue device acts as fake DHCP server
- Assigns wrong IP/gateway → traffic interception

MAC Flooding

- Switch CAM table overflow
- Switch behaves like hub → traffic sniffing

STP Attack (BPDU Manipulation)

- Attacker becomes root bridge
- Controls traffic flow or causes disruption

Mitigation Techniques

VLAN Segmentation + ACLs:

- Isolate departments (10, 20, 30)
- Block Guest access to internal VLANs

Port Security:

- Limit MAC addresses per port
- Prevent unauthorized devices

DHCP Snooping (Recommended)

- Allows DHCP only from trusted ports
- Blocks rogue DHCP servers

Dynamic ARP Inspection (DAI)

- Validates ARP packets
- Prevents ARP spoofing

6. Testing, Verification & Live Demo

DHCP Verification

```
Vlan30_GUEST> ip dhcp
DDORA IP 10.0.30.3/24 GW 10.0.30.1

Vlan30_GUEST> show ip

NAME       : Vlan30_GUEST[1]
IP/MASK    : 10.0.30.3/24
GATEWAY    : 10.0.30.1
DNS        : 8.8.8.8
DHCP SERVER : 10.0.30.1
DHCP LEASE  : 86392, 86400/43200/75600
MAC        : 00:50:79:66:68:01
LPORT      : 20043
RHOST:PORT  : 127.0.0.1:20044
MTU        : 1500

Vlan30_GUEST>
```

Management Access

```
Vlan10_MGMT> ping 10.0.20.10

84 bytes from 10.0.20.10 icmp_seq=1 ttl=63 time=30.109 ms
84 bytes from 10.0.20.10 icmp_seq=2 ttl=63 time=18.756 ms
84 bytes from 10.0.20.10 icmp_seq=3 ttl=63 time=18.059 ms
84 bytes from 10.0.20.10 icmp_seq=4 ttl=63 time=18.442 ms
84 bytes from 10.0.20.10 icmp_seq=5 ttl=63 time=16.037 ms

Vlan10_MGMT> ping 10.0.30.10

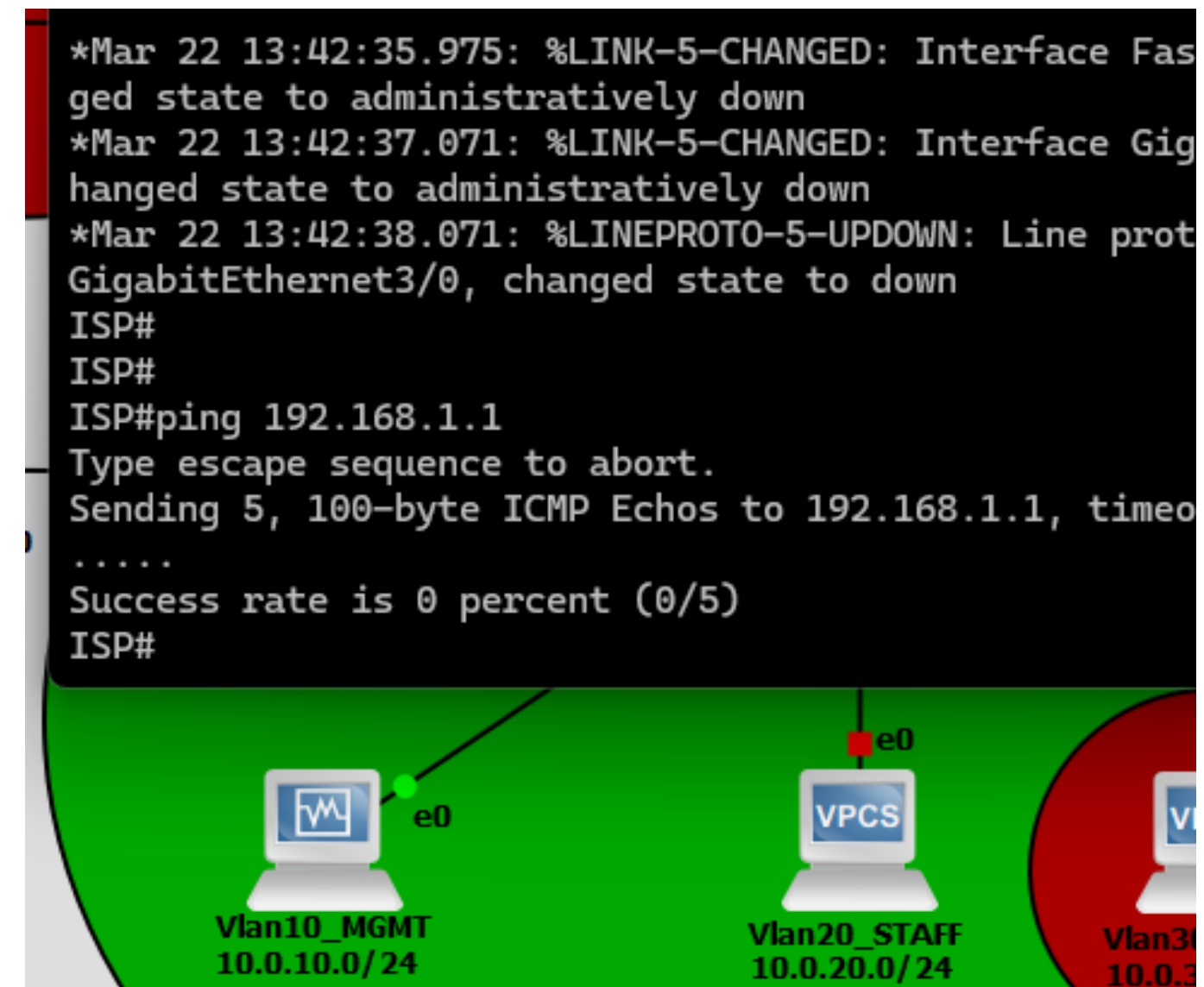
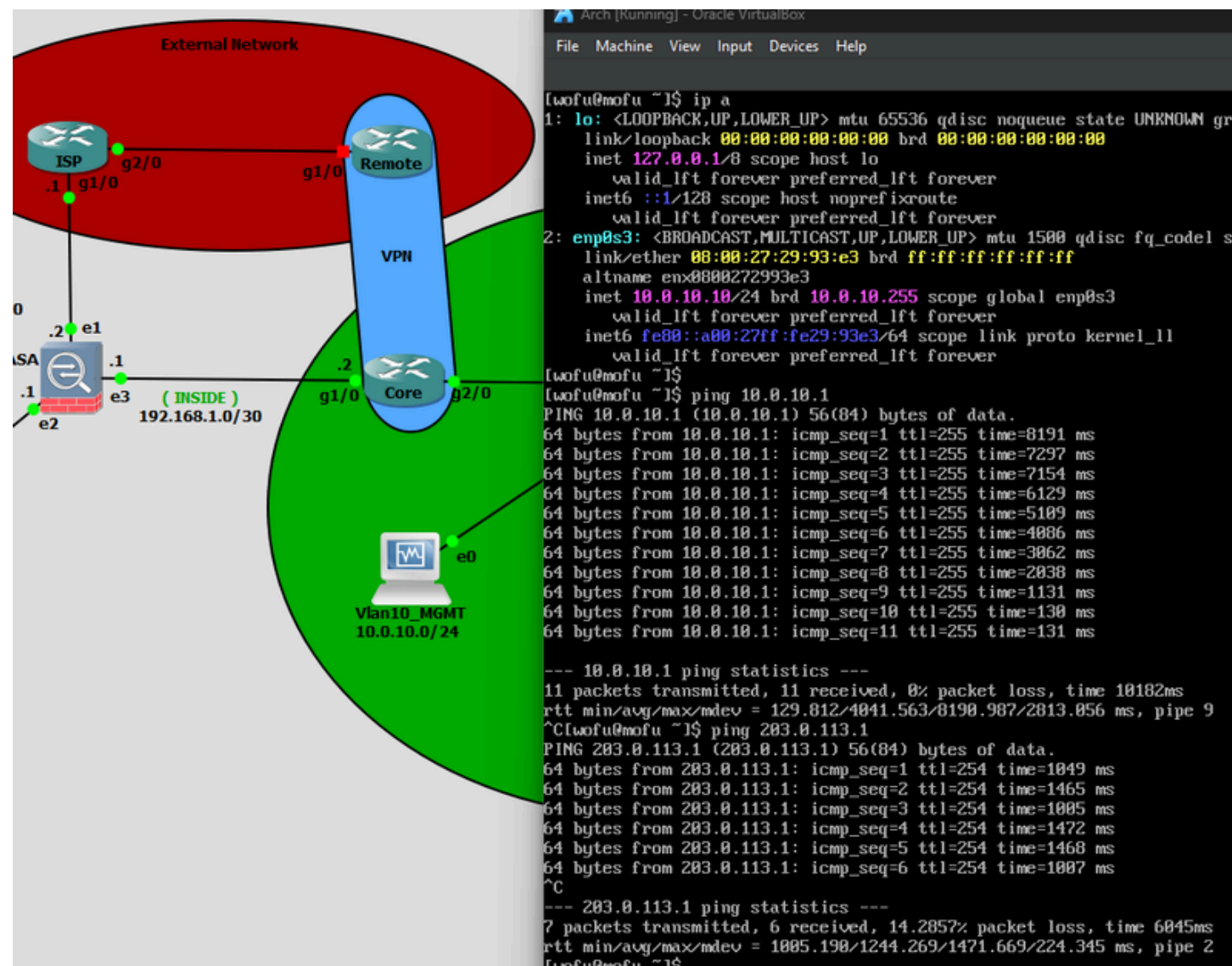
10.0.30.10 icmp_seq=1 timeout
10.0.30.10 icmp_seq=2 timeout
10.0.30.10 icmp_seq=3 timeout
10.0.30.10 icmp_seq=4 timeout
10.0.30.10 icmp_seq=5 timeout

Vlan10_MGMT> |
```

6. Testing, Verification & Live Demo (cont.)

VLAN - Outside Network

Outside - Inside



6. Testing, Verification & Live Demo (cont.)

VPN Tunnel

```
Vlan20_STAFF>
Vlan20_STAFF>
Vlan20_STAFF> ping 192.168.100.1

host (10.0.20.1) not reachable

Vlan20_STAFF> ping 192.168.100.1

84 bytes from 192.168.100.1 icmp_seq=1 ttl=255 time=995.602 ms
84 bytes from 192.168.100.1 icmp_seq=2 ttl=255 time=23.983 ms
192.168.100.1 icmp_seq=3 timeout
84 bytes from 192.168.100.1 icmp_seq=4 ttl=255 time=996.271 ms
84 bytes from 192.168.100.1 icmp_seq=5 ttl=255 time=49.922 ms

Vlan20_STAFF>
```

SSH From VLAN10 to ASA

```
[wofu@mofu ~]$ ssh -o KexAlgorithms=+diffie-hellman-group1-sha1 -o HostKeyAlgorithms=+ss
admin@192.168.1.1's password:
User admin logged in to ciscoasa
Logins over the last 1 days: 4. Last login: 12:52:49 UTC Mar 22 2026 from 10.0.10.10
Failed logins since the last login: 0.
Type help or '?' for a list of available commands.
ciscoasa> en
Password:
ciscoasa#
ciscoasa#
ciscoasa# conf t
ciscoasa(config)# exit
ciscoasa# show int ip bri
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 203.0.113.2 YES CONFIG up up
GigabitEthernet0/1 172.16.1.1 YES CONFIG up up
GigabitEthernet0/2 192.168.1.1 YES CONFIG up up
Management0/0 unassigned YES unset administratively down up
ciscoasa#
```

7. Project Conclusion

Network Segmentation

Isolate guest traffic from internal corporate. ACLs to prevent unauthorized access.

Management Plane Protection

Restrict SSH and administrative access strictly to the internal management network

Layer 2 Switch Hardening

Implement Port Security and DHCP Snooping on access switches to prevent unauthorized DHCP.

Thank You
we are open to
Q&A